



**COMSATS University
Islamabad
Abbottabad Campus**

Department of Computer Science

SUBJECT:

SOFTWARE QUALITY ENGINEERING

ASSIGNMENT 3.

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1. CodeGen with Playwright

Introduction

Playwright is an open-source automation testing framework developed by Microsoft. It is used to test modern web applications across different browsers such as Chromium, Firefox, and WebKit. One of its key features is **CodeGen**, which automatically generates test scripts by recording user actions.

History

Playwright was introduced by Microsoft in **2020** to provide faster and more reliable browser automation. It was designed to support modern web applications and multiple browsers. CodeGen was later introduced to simplify the process of creating automation scripts.

What is CodeGen?

CodeGen is a built-in recording tool in Playwright that captures user interactions with a website. It converts actions such as clicking, typing, and selecting options into executable test scripts. This reduces the need for manual coding.

How CodeGen Works

1. Install Playwright and open the CodeGen tool.
2. Enter the website URL.
3. Perform actions on the website.
4. Playwright records the actions and generates the automation script automatically.

Requirements

- Node.js installed on the system.
- Playwright package installed.
- Supported browser (Chromium, Firefox, or WebKit).
- Code editor such as Visual Studio Code.
- Basic knowledge of automation testing.

Features

- Automatically records user actions.
- Generates automation scripts instantly.
- Supports Chromium, Firefox, and WebKit browsers.
- Supports JavaScript, TypeScript, Python, Java, and C#.

- Generates reliable element locators automatically.
- Easy to edit, reuse, and maintain test scripts.
- Integrates with CI/CD pipelines.
- Supports modern web applications and dynamic web elements.

Uses of CodeGen with Playwright

CodeGen is widely used to create automation test scripts without writing code manually. It helps automate common tasks such as login, registration, search, checkout, and form validation. It is also useful for regression testing because the generated scripts can be reused whenever the application is updated. Developers and testers use CodeGen to quickly build automation tests for modern web applications.

Supported Programming Languages

Playwright CodeGen supports multiple programming languages, making it flexible for different projects. These include **JavaScript, TypeScript, Python, Java, and C#**. Developers can choose the language according to their project requirements.

Advantages

- Saves time by generating scripts automatically.
- Reduces manual coding effort.
- Minimizes human errors during test creation.
- Supports cross-browser and cross-platform testing.
- Improves testing speed and accuracy.
- Makes automation easier for beginners.
- Increases test coverage and software reliability.
- Supports continuous testing in Agile and DevOps projects.

Productivity Benefits

CodeGen significantly improves the productivity of software testing teams. It reduces the time required to create automation scripts and allows testers to focus on complex testing scenarios instead of repetitive tasks. Since scripts are generated automatically, testing becomes faster, more accurate, and easier to maintain. This helps organizations deliver high-quality software within shorter development cycles.

Applications of CodeGen

CodeGen is used in many real-world software projects for automating functional testing, regression testing, smoke testing, and user interface (UI) testing. It is also used in Agile and DevOps environments where applications are updated frequently. The generated scripts can be integrated with Continuous Integration and Continuous Delivery (CI/CD) pipelines for continuous automated testing.

Limitations

- Complex test cases may need manual editing.
- Dynamic web elements may require custom locators.
- Scripts need updates when the user interface changes.
- Advanced testing scenarios cannot always be recorded automatically.

Example

Suppose a tester wants to automate the login page of an online shopping website. The tester opens CodeGen, enters the website URL, types the login credentials, and clicks the **Login** button. Playwright records these actions and automatically generates the complete automation script.

Importance in Software Quality Engineering

Playwright CodeGen is an important tool in Software Quality Engineering because it simplifies automation testing and improves software quality. It reduces manual effort, increases testing efficiency, and helps detect defects early in the development process. It also supports continuous integration, continuous testing, and faster software delivery. By automating repetitive testing tasks, CodeGen allows Quality Engineers to focus on improving the overall reliability and performance of software applications.

2. Use of MCP in Software Quality Engineering (SQE)

Introduction

Model Context Protocol (MCP) is an open standard that allows Artificial Intelligence (AI) systems to connect securely with software development tools, databases, APIs, and project documentation. It helps AI understand the complete project context before performing any task. This makes AI assistance more accurate, efficient, and useful in Software Quality Engineering.

What is MCP?

MCP (Model Context Protocol) acts as a communication bridge between AI applications and external software tools. It allows AI to access project files, source code, documentation, test reports, and development resources securely. By using real project information, AI can provide context-aware and reliable responses instead of relying only on user prompts.

History

MCP was introduced to solve the problem of AI working with limited information. As AI became more common in software development, there was a need for a standard protocol that could connect AI with external tools and services. Today, MCP is becoming an important technology in AI-assisted software engineering.

How MCP Works

1. The user sends a request to the AI assistant.
2. MCP securely connects the AI with external tools and resources.
3. The required project information is retrieved.
4. The AI analyzes the information and performs the requested task.
5. The results are returned to the user.

Components of MCP

- **AI Client:** Receives user requests and communicates with the MCP server.
- **MCP Server:** Connects AI with external resources and manages requests.
- **Tools and Services:** Includes IDEs, Git repositories, databases, APIs, documentation, and testing tools.
- **Communication Protocol:** Ensures secure and standardized data exchange.

Uses of MCP in Software Quality Engineering

MCP is widely used in Software Quality Engineering to improve software testing and development. It helps generate accurate test cases, review source code, detect software bugs, and analyze project documentation. It also supports requirement traceability, evaluates test reports, and assists in Continuous Integration (CI) and Continuous Testing. By providing AI with real project information, MCP makes software quality assurance more efficient and reliable.

Features of MCP

- Secure communication between AI and software tools.
- Access to real project files and documentation.
- Integration with APIs, databases, and Git repositories.
- Supports AI-assisted testing and code analysis.
- Improves collaboration between developers, testers, and AI systems.

Advantages of MCP

- Improves the accuracy of AI-generated results.
- Reduces manual testing and documentation effort.
- Speeds up bug detection and code review.
- Enhances collaboration among development teams.
- Supports faster software development and delivery.
- Improves the overall quality and reliability of software.

Challenges

- Data privacy and security must be maintained.
- Integration with multiple tools can be complex.
- Proper authentication and access permissions are required.
- Performance depends on connected tools and services.
- Organizations must ensure secure handling of project data.

Example

Suppose a software tester asks an AI assistant to generate functional test cases for a login module. Using MCP, the AI securely accesses the Software Requirement Specification (SRS), project documentation, and existing test data. Based on this information, it creates accurate and relevant test cases that match the actual project requirements.

Importance in Software Quality Engineering

MCP plays an important role in modern Software Quality Engineering by enabling AI to work with real project data instead of assumptions. It supports automated testing, code analysis, documentation review, and bug detection while reducing manual effort. As AI continues to evolve, MCP will become an essential technology for improving software quality, increasing productivity, and delivering reliable software products.

3. Google Antigravity

Introduction

Google Antigravity is a fun JavaScript-based web experiment that simulates gravity on the Google homepage. When the page loads, elements such as the logo, search bar, and buttons fall to the bottom of the screen. It demonstrates how JavaScript can create dynamic and interactive web pages using animations and browser rendering techniques.

History

Google Antigravity was created by **Ricardo Cabello (Mr.doob)** as a web experiment using JavaScript. Although many people think it is an official Google feature, it was developed for learning and entertainment purposes. It became popular because of its creative design and realistic gravity simulation.

What is Google Antigravity?

Google Antigravity is an interactive webpage where gravity is applied to all webpage elements. Instead of remaining fixed, objects fall naturally and can be dragged and moved using the mouse. It demonstrates browser animations, event handling, and physics-based effects in web applications.

How Google Antigravity Works

1. The webpage loads normally.
2. JavaScript removes the fixed positions of webpage elements.
3. Gravity is applied, causing all objects to fall.
4. Users can drag, drop, and move the elements freely.
5. The browser continuously updates the position of the objects to create a realistic animation.

Technologies Used

Google Antigravity is developed using **HTML, CSS, and JavaScript**. It also uses **DOM manipulation**, browser rendering, and animation techniques to simulate gravity and user interaction. These technologies make the webpage interactive and visually engaging.

Features

- Simulates gravity on webpage elements.
- Supports drag-and-drop interaction.
- Uses real-time JavaScript animations.
- Demonstrates dynamic webpage behavior.
- Runs directly in a web browser without additional software.
- Provides an interactive learning experience.

How Google Antigravity Can Be Utilized in SQE for QA and Software Testing

Although Google Antigravity is not a testing tool, it can be used as a sample web application for Quality Assurance (QA) testing. Software testers can verify browser compatibility, user interface (UI) behavior, drag-and-drop functionality, animation performance, and page responsiveness. It

also helps evaluate JavaScript execution and identify rendering issues across different browsers and devices.

Advantages for QA and Testing

- Helps test browser compatibility across multiple browsers.
- Verifies JavaScript functionality and UI interactions.
- Tests animation performance and responsiveness.
- Supports usability and user experience (UX) testing.
- Helps identify rendering and interface issues.
- Improves the quality of interactive web applications.

Limitations

- It is not an official Google product.
- It is designed mainly for learning and entertainment.
- It has no practical use for web searching.
- It cannot replace professional software testing tools.
- It is not suitable for production websites.

Example

Suppose a QA engineer wants to verify whether animations and drag-and-drop features work correctly on Chrome, Firefox, and Edge. A project similar to Google Antigravity can be used to test these interactive features. If animations behave differently on different browsers, the tester can identify and fix compatibility issues before releasing the software.

Importance in Software Quality Engineering

Google Antigravity helps Software Quality Engineers understand how interactive web applications behave in different environments. It is useful for testing browser compatibility, animation performance, user interface responsiveness, and usability. By applying these testing techniques, QA teams can improve software quality, enhance the user experience, and ensure that web applications perform consistently across different platforms.

Contribution of group members:

Student Name	Registration No.	Contribution	Topics Covered

Aiman Shabbir	FA22-BSE-104	50%	CodeGen with Playwright • Introduction • History • What is CodeGen? • How CodeGen Works • Requirements • Features • Uses of CodeGen with Playwright • Supported Programming Languages • Advantages • Productivity Benefits • Applications of CodeGen • Limitations • Example • Importance in Software Quality Engineering Use of MCP in Software Quality Engineering (SQE) • Introduction • What is MCP? • History • How MCP Works
Hamza Ali	FA23-BSE-142	50%	• Components of MCP • Uses of MCP in Software Quality Engineering • Features • Advantages • Challenges • Example • Importance in Software Quality Engineering Google Antigravity • Introduction • History • What is Google Antigravity? • How Google Antigravity Works • Technologies Used • Features • How Google Antigravity Can Be Utilized in

			SQE for QA and Software Testing • Advantages for QA and Testing • Limitations • Example • Importance in Software Quality Engineering
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